

Christopher Musco

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Education

Massachusetts Institute of Technology

Ph.D. Candidate, Computer Science – Theory of Computation

Advisor: Jonathan Kelner

Masters Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

Cambridge, MA

2013 – present

Yale University

B.S. Applied Mathematics, B.S. Computer Science

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (with Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (with Daniel Spielman)

New Haven, CT

2008 – 2012

Other Academic Positions

Yahoo Labs

Research Intern, Scalable Machine Learning Group

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on theoretical matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

New York, NY

Summer 2015

Research Areas

theory of algorithms ◦ scalable machine learning ◦ randomized linear algebra ◦ low rank approximation
dimensionality reduction ◦ sketching and streaming ◦ algorithmic graph theory ◦ iterative matrix algorithms

Publications

Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.

Conference Papers.....

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *SODA* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *ALLENEX* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *ICML* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *NIPS* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers).

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. Invited to SICOMP Special Issue.

Other.....
Provably Useful Kernel Matrix Approximation in Linear Time. Cameron Musco, Christopher Musco. *In submission.*

Teaching and Mentorship

Massachusetts Institute of Technology

Teaching Assistant, Advanced Algorithms (6.854/18.415) 2016

Held weekly office hours, wrote problem sets, and compiled notes and resources for an updated version of this capstone graduate algorithms course. Wrote guide on topics and expectations for student projects. Prepared and delivered a lecture on linear programming relaxations.

Workshop Leader, Technical Communication Skills for Graduate Students (6.S977) 2016

Lead weekly workshops that supplemented lecture material with hands-on writing and presentation exercises. Met with student one-on-one outside of class to offer individual support and coaching in technical communication.

Communication Advisor 2015 – present

Helping to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offer feedback, guidance, and coaching to individual students during weekly office hours (40+ individual appointments to date).

Research Science Institute at MIT

Summer Research Advisor 2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Chris' project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Yale University

Teaching Assistant, Introduction to Computer Science 2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Teaching Assistant, Introduction to Programming 2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

Grader, Introductory Calculus 2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Talks

Introduction to Linear Sketching

MIT, Theory Retreat Sept. 2016

Principal Component Projection without Principal Component Analysis

International Conference on Machine Learning (ICML) June 2016

Ridge Leverage Score Sampling

National Institute of Informatics, Shonan Meeting July 2016

MIT, Algorithms & Complexity Seminar July 2016

University of Utah, Data Group Meeting Jan. 2016

Randomized Block Krylov Methods

Neural Information Processing Systems (NIPS) Poster Session Dec. 2015

Large Scale Column Subset Selection

Yahoo Labs, Science Week Poster Session July 2015

Dimensionality Reduction for K-Means Clustering

IBM T.J. Watson, Mathematical Sciences Research Seminar Aug. 2015

Symposium on Theory of Computing (STOC)	June 2015
MIT, Theory Lunch	June 2014
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency Site Visit	Mar. 2015
Network and Distributed System Security Symposium (NDSS)	Feb. 2015
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning Research Seminar	June 2015
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day Poster Session	Apr. 2015
Harvard University, Theory Seminar	Nov. 2014
Foundations of Computer Science (FOCS)	Oct. 2014

Service

External Reviewer.....	
SIAM Journal on Matrix Analysis and Applications (SIMAX)	2017
IEEE Transactions on Knowledge and Data Engineering (TKDE)	2017
Conference on Learning Theory (COLT)	2016
Symposium on Discrete Algorithms (SODA)	2016, 2017
IEEE Transactions on Signal Processing (TSP)	2016, 2017
Neural Information Processing Systems (NIPS)	2015, 2016
European Symposia on Algorithms (ESA)	2015
Foundations of Computer Science (FOCS)	2015
International Colloquium on Automata, Languages and Programming (ICALP)	2015
Network and Distributed System Security Symposium (NDSS)	2015

Honors and Awards

National Science Foundation	2014
<i>Graduate Research Fellowship.</i>	
Yale University	2012
<i>Cum Laude, Distinction in Both Majors.</i>	

Professional Experience

Redfin	Seattle, WA
<i>Software Engineer, Data Team</i>	2012 – 2014
Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.	
Elysium Digital	Cambridge, MA
<i>Summer Consultant</i>	Summer 2011
Provided technical expertise in software patent cases at leading litigation consulting company.	
Amicus	New Haven, CT
<i>Software Developer</i>	2010 – 2011
Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.	

Moody's Investors Service**New York, NY***Intern, Asset Finance Group**Summer 2009*

Established and monitored bond ratings on securitizations in three asset classes. Wrote code to automate analysis for student loan data and to compile records on aircraft leases.

U.S. Department of Agriculture**East Greenwich, RI***Intern, Natural Resources Conservation Service**Summer 2007*

Collected and analyzed data for two major state conservation and research projects. Integrated mapping software and database tools to develop a system for compiling and presenting soil data.